

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 1) - Single Correct

Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R :

Assertion A: Size of Bk^{3+} ion is less than Np^{3+} ion. Reason R : The above is a consequence of the lanthanoid contraction. In the light of the above statements, choose the correct answer from the options given below :

- (1) A is false but R is true
- (2) Both A and R are true but R is not the correct explanation of A
- (3) Both A and R are true and R is the correct explanation of A
- (4) A is true but R is false

Q2: 16 March (Shift 1) - Single Correct

Given below are two statements:

Statement I : The E° value of $\text{Ce}^{4+}/\text{Ce}^{3+}$ is $+1.74 \text{ V}$

Statement II : Ce is more stable in Ce^{4+} state than Ce^{3+} state.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both statement I and statement II are correct
- (2) Statement I is incorrect but statement II is correct
- (3) Both statement I and statement II are incorrect
- (4) Statement I is correct but statement II is incorrect.

Q3: 16 March (Shift 2) - Numerical

When 35 mL of 0.15M lead nitrate solution is mixed with 20 mL of 0.12M chromic sulphate solution, $\times 10^{-5}$ moles of lead sulphate precipitate out. (Round off to the Nearest Integer).

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Q4: 17 March (Shift 1) - Single Correct

Given below are two statements:

Statement I : Potassium permanganate on heating at 573 K forms potassium manganate.

Statement II : Both potassium permanganate and potassium manganate are tetrahedral and paramagnetic in nature.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is true but statement II is false
- (2) Both statement I and statement II are true
- (3) Statement I is false but statement II is true
- (4) Both statement I and statement II are false

Q5: 17 March (Shift 2) - Single Correct

The common positive oxidation states for an element with atomic number 24, are :

- (1) +2 to +6
- (2) +1 and +3 to +6
- (3) +1 and +3
- (4) +1 to +6

Q6: 17 March (Shift 2) - Numerical

On complete reaction of FeCl_3 with oxalic acid in aqueous solution containing KOH, resulted in the formation of product A. The secondary valency of Fe in the product A is _____

Answer Key**Q1 (4)****Q2 (4)****Q3 (525)****Q4 (1)****Q5 (1)****Q6 (6)**

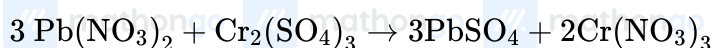
Q1 (4)

Size of ${}_{97}\text{Bk}^{3+}$ ion is less than that of ${}_{93}\text{Np}^{3+}$ due to actinoid contraction. As we know that in a period from left to right ionic radius decreases and in actinide series it is due to actinoid contraction.

Q2 (4)

The E° value for $\text{Ce}^{4+}/\text{Ce}^{3+}$ is +1.74 V because the most stable oxidation state of lanthanide series elements is +3. It means Ce^{3+} is more stable than Ce^{4+} .

Q3 (525)



35ml 20ml

0.15M 0.12M

= 5.25 m. mol = 2.4 m. mol 5.25 m. mol

= 5.25×10^{-3} moltherefore moles of PbSO_4 formed = $5.25 \times 10^{-3} = 525 \times 10^{-5}$

Q4 (1)

Potassium
permanganatePotassium
manganate

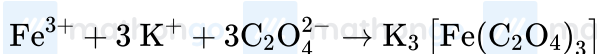
Statement-I is correct.

Statement-II is incorrect.

Q5 (1)

 $\text{Cr}(Z = 24)$ $[\text{Ar}] 4s^1 3d^5$ Cr shows common oxidation states starting from +2 to +6.

Q6 (6)



(A) Secondary valency of Fe in 'A' is 6.